EXPERIMENT 11: EXAMINE NETWORK ADDRESS TRANSLATION USING CISCO PACKET TRACER

AIM:

To configure and examine Network Address Translation (NAT) using Cisco Packet Tracer.

PROCEDURE:

1. Build a topology with an inside LAN, a router and an outside network or server.
2. Configure the inside and outside router interfaces with IP addresses.
3. Configure the PCs with private IP addresses and the correct default gateway.
4. Mark the router interfaces as NAT inside and NAT outside.
5. Configure a standard ACL for the inside subnet and enable NAT overload.
6. Generate traffic from an inside PC and inspect the translation table.
7. Verify that the inside and outside interfaces are operational before testing NAT.
8. Generate traffic from an inside host to the outside network to create a translation entry.
9. Use the NAT statistics and translation commands to inspect the mapping.
10. If no translation appears, check the access list, interface roles, routing and host default gateway.

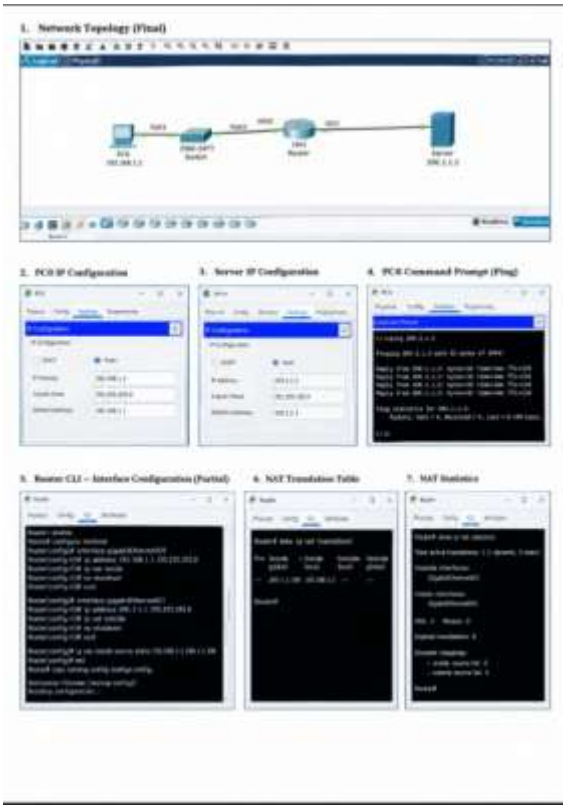
CODE:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# ip nat inside
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitEthernet 0/1
```

```
Router(config-if)# ip address 203.0.113.2 255.255.255.0
Router(config-if)# ip nat outside
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# access-list 1 permit 192.168.1.0
0.0.0.255
Router(config)# ip nat inside source list 1 interface
gigabitEthernet 0/1 overload
Router(config)# end
Router# show ip nat translations
Router# show ip nat statistics

Note: Interface names must match the router model in
Packet Tracer.
```

OUTPUT:



RESULT:

Thus, NAT overload was configured and the translation table was examined in Cisco Packet Tracer